Cloud Suryalog — Get Datalog API Integration SOP

This Standard Operating Procedure (SOP) governs the technical integration, authentication protocols, and schema reference for the Cloud Suryalog **Get Datalog API**. External applications, SCADA aggregators, ERPs, and monitoring dashboards utilize this interface to extract authenticated telemetry comprising Weather Monitoring Stations (WMS), Energy Meters, Power Inverters, and String Monitoring Boxes (SMB).

1. API ARCHITECTURE & DYNAMIC VERSION ENDPOINT

Production URL Template:
`https://cloud.suryalog.ae/api/get_datalog_{version}.php`

HTTP Method: `POST`
Content-Type: `application/json`

Dynamic Versioning Specification: The `{version}` path component must be specified dynamically in your system integration pipeline (e.g., v7, v7.1, v8, or future releases). Do not hardcode static versions in production scripts without parameterization.

2. ENTERPRISE SECURITY, CREDENTIALS & IP PROTECTION

Security Mandate: The API `secret` is a private cryptographic token. It must never be published in client-side repositories, browser scripts, or mobile codebases. Ingesting systems must route calls via a secure server-side proxy with TLS 1.2+ encryption.

3. HTTP POST REQUEST PAYLOAD SPECIFICATION

```
{ "secret": "cd8ae85e44e2ca7d997904893f8fb7fe", "plant": "[ENCRYPTED_PLANT_IDENTIFIER]", "format": "std", "for": "data", "stime": "1776844220", "etime": "1776844220" }
```

PARAMETER	TYPE	REQUIRED	TECHNICAL SPECIFICATION & BUSINESS PURPOSE
`secret`	String	Yes	Unique 32-character hexadecimal cryptographic API authentication key provided by Suryalogix.
`plant`	String	Yes	Encrypted plant identification string mapping directly to the physical installation in Suryalog Cloud.
`format`	String	Yes	Response formatting layout standard. Must be set to `std` (Standard).
`for`	String	Yes	Query operation scope filter: use `data` for raw time-series logs; use `report` for summarized datasets.
`stime`	String	Yes	Start range timestamp represented as a standard 10-digit Unix epoch timestamp (UTC/Site local epoch).
`etime`	String	Yes	End range timestamp represented as a standard 10-digit Unix epoch timestamp (UTC/Site local epoch).

4. RESPONSE ENVELOPE HIERARCHY & JSON TREE

Top-Level Response Schema:


```
{ "plant": "EncryptedPlantID", "server_time": 1776319998, "result": 0, "cmsg": "Success", "data": { ... } }
```


Telemetry Tree Structure:

```
data └── {TIMESTAMP} ├── weather | └── [Weather Object] ├── meter | └── SM ├── inverter | └── I1 ... In └── SMB └── SMB1 ... SMBn
```

5. WEATHER MONITORING STATION (WMS) — COMPLETE FIELD DICTIONARY

The Weather Station (WS) object delivers all meteorological, thermal, and insolation data vital for Performance Ratio (PR) computations, IEC 61724 normalization, and irradiance yield analysis.

Object Path: `data → {TIMESTAMP} → weather`

PARAMETER KEY	UNIT	DATA TYPE	FIELD ENGINEERING DESCRIPTION & USAGE
weather_type	`String`	String	Identifies the weather telemetry profile (standard: WS for Weather Station).
W2_val	`W/m²`	Float (String)	Instantaneous solar irradiance measured by the primary pyranometer (POA or GHI).
W2_energy	`kWh/m²`	Float (String)	Cumulative solar irradiation energy logged over the active measurement slice.
W3_val	`°C`	Float (String)	Instantaneous module surface temperature from reference PV back-surface RTD sensor.
W3_energy	`Numeric`	Float (String)	Accumulator register / reserve energy coefficient for the module temperature channel.
W4_val	`°C`	Float (String)	Instantaneous ambient air temperature measured inside a naturally aspirated radiation shield.
W4_energy	`Numeric`	Float (String)	Accumulator register / reserve energy coefficient for ambient temperature channel.
W2_DAY_MIN	`W/m²`	Float (String)	Minimum instantaneous solar irradiance logged since midnight of the active diurnal cycle.
W2_DAY__MAX	`W/m²`	Float (String)	Peak maximum solar irradiance logged during daylight operation today.
W2_DAY_AVG	`W/m²`	Float (String)	Arithmetic average solar irradiance logged across daylight operating hours today.
W2_INS	`kWh/m²`	Float (String)	Current day cumulative solar insolation received on the sensor surface.
YESTW2_INS	`kWh/m²`	Float (String)	Total cumulative solar insolation logged across the full 24 hours of yesterday.
MONTH_W2_INS	`kWh/m²`	Float (String)	Cumulative solar insolation accumulated from Day 1 to current timestamp of this month.

6. ENERGY METER PARAMETERS & UNITS (PATH: DATA → {TIMESTAMP} → METER → SM)

Comprehensive electrical measurements captured from the plant main revenue/check meters (MFM), including voltage harmonics, phase currents, and instantaneous powers.

PARAMETER KEY	UNIT	DATA TYPE	ENGINEERING DESCRIPTION & SIGNAL MEANING
meter_type	String	String	Installed energy meter model designation and protocol configuration.
suralog_status	Status	Integer/String	Data logger physical communication link status (1 = Active/Healthy, 0 = Offline).
user_status	Status	Integer/String	Customer account operating status and authorization state.
meter_addon_data	Raw/JSON	String	Supplemental diagnostic information or vendor proprietary meter register payload.
meter_addon_active	Boolean	Integer/String	Indicates whether meter addon extensions are enabled (1 = True, 0 = False).
meter_data_active	Boolean	Integer/String	Indicates active real-time transmission of electrical instantaneous metrics.
meter_energy_active	Boolean	Integer/String	Indicates active transmission of cumulative energy counter registers.
VLL	V	Float (String)	Average three-phase line-to-line AC voltage.
VLN	V	Float (String)	Average three-phase line-to-neutral AC voltage.
V12	V	Float (String)	Instantaneous AC voltage between Phase 1 and Phase 2.
V23	V	Float (String)	Instantaneous AC voltage between Phase 2 and Phase 3.
V31	V	Float (String)	Instantaneous AC voltage between Phase 3 and Phase 1.
V1	V	Float (String)	Phase 1 to neutral AC voltage (L1-N).
V2	V	Float (String)	Phase 2 to neutral AC voltage (L2-N).
V3	V	Float (String)	Phase 3 to neutral AC voltage (L3-N).
A1 / I1	A	Float (String)	Phase 1 root-mean-square (RMS) current.
A2 / I2	A	Float (String)	Phase 2 root-mean-square (RMS) current.
A3 / I3	A	Float (String)	Phase 3 root-mean-square (RMS) current.
ILN / Inutral	A	Float (String)	Measured or computed neutral conductor current.
FREQ	Hz	Float (String)	AC grid frequency measured at the meter point of coupling.
PF1, PF2, PF3	Factor	Float (String)	Individual displacement power factor for Phase 1, Phase 2, and Phase 3.
PFT	Factor	Float (String)	Total aggregated three-phase system power factor.

6. ENERGY METER (CONTINUED) — POWERS, ACCUMULATORS & PERFORMANCE RATIOS

PARAMETER KEY	UNIT	DATA TYPE	ENGINEERING DESCRIPTION & ANALYTICAL PURPOSE
W1, W2, W3	kW	Float (String)	Active power generated or consumed across Phase 1, Phase 2, and Phase 3.
WT	kW	Float (String)	Total three-phase instantaneous active power.
VA1, VA2, VA3	kVA	Float (String)	Apparent power delivered or received across Phase 1, Phase 2, and Phase 3.
VAT	kVA	Float (String)	Total three-phase instantaneous apparent power.
VAR1, VAR2, VAR3	kVAR	Float (String)	Reactive power logged per phase (L1, L2, L3).
VART	kVAR	Float (String)	Total three-phase instantaneous reactive power.
WHImp / WHExp	Wh	Float (String)	Total cumulative active imported energy (from grid) and exported energy (to grid).
WHTot	Wh	Float (String)	Total net cumulative active energy recorded on the meter register.
VAHImp / VAHExp	VAh	Float (String)	Cumulative apparent imported energy and apparent exported energy registers.
VAHTot	VAh	Float (String)	Total net cumulative apparent energy register.
VARHiImp / VARHiExp	VARh	Float (String)	Cumulative inductive reactive imported energy and inductive exported energy.
VARHcImp / VARHcExp	VARh	Float (String)	Cumulative capacitive reactive imported energy and capacitive exported energy.
VARHTot	VARh	Float (String)	Total net cumulative reactive energy counter.
AC_CAP / DC_CAP	kW / kWp	Float (String)	Contractual plant ratings: installed AC inverter capacity and peak DC array capacity.
plant_slf	kWh	Float (String)	Plant internal auxiliary self-consumption energy.
DAY_MIN / MAX / AVG	kW	Float (String)	Daily minimum, peak maximum, and mean active power recorded today.
KWH_Day / IMP_Day / EXP_Day	kWh	Float (String)	Today's cumulative generated energy, imported grid energy, and exported energy.
YEST_KWH / IMP / EXP	kWh	Float (String)	Yesterday's total generated energy, imported grid energy, and exported energy.
MONTH_KWH / IMP / EXP	kWh	Float (String)	Current month-to-date total generation, grid import, and grid export energy.
YEAR_KWH / IMP / EXP	kWh	Float (String)	Current year-to-date total generation, grid import, and grid export energy.
Day_PR / YEST_PR / MONTH_PR	%	Float (String)	Plant Performance Ratio calculated for today, yesterday, and active calendar month.
Day_CUF_AC / Day_CUF_DC	%	Float (String)	Daily Capacity Utilization Factor calculated against AC capacity and DC peak capacity.
YEST_CUF_AC / DC	%	Float (String)	Yesterday's Capacity Utilization Factor referenced against AC and DC ratings.
MONTH_CUF_AC / DC / days	% / Days	Float / Int	Monthly AC/DC Capacity Utilization Factors and count of active calculation days.

7. SOLAR INVERTER TELEMETRY DICTIONARY (PATH: DATA → {TIMESTAMP} → INVERTER → I1...IN)

High-granularity operational telemetry captured across string and central inverters, indexing DC inputs, AC conversion states, heatsink temperatures, and curtailment limits.

PARAMETER KEY	UNIT	DATA TYPE	ENGINEERING DESCRIPTION & CONTROL MAPPING
`inverter_type`	`String`	String	OEM inverter model identity string, protocol driver, or firmware signature.
`suryalog_status`	`Status`	Integer/String	Inverter-to-logger RS485 / Modbus TCP communication link status.
`user_status`	`Status`	Integer/String	Inverter operating state (e.g., Running, Standby, Grid Fault, Derating).
`inverter_addon_data`	`Raw/JSON`	String	Additional telemetry payload or extended manufacturer diagnostic error registers.
`inverter_addon_active`	`Boolean`	Integer/String	State of extended feature registers for advanced analytics.
`inverter_data_active`	`Boolean`	Integer/String	Indicates active real-time transmission of inverter instantaneous telemetry.
`inverter_energy_active`	`Boolean`	Integer/String	Indicates active transmission of inverter energy accumulation counters.
`DC_V`	`V`	Float (String)	Operating direct current (DC) bus voltage at inverter input stages.
`DC_I`	`A`	Float (String)	Operating direct current (DC) input array current.
`DC_W`	`W / kW`	Float (String)	Total direct current (DC) input power fed from the photovoltaic array.
`VT`	`V`	Float (String)	Three-phase alternating current (AC) output grid terminal voltage.
`IT`	`A`	Float (String)	Total combined alternating current (AC) output current delivered to the bus.
`WT`	`W / kW`	Float (String)	Total three-phase active power (AC) exported by the inverter.
`W1, W2, W3`	`W / kW`	Float (String)	Instantaneous AC active power generated across Phase 1, Phase 2, and Phase 3.
`VAT`	`VA / kVA`	Float (String)	Total apparent AC output power at inverter terminals.
`VART`	`VAR / kVAR`	Float (String)	Total reactive AC power supplied or absorbed by the inverter.
`PFT`	`Factor`	Float (String)	Operating power factor at the inverter AC connection.
`FREQ`	`Hz`	Float (String)	Synchronized grid frequency detected by the inverter PLL.
`TEMP_INT`	`°C`	Float (String)	Internal inverter enclosure cabinet ambient temperature.
`TEMP_HS`	`°C`	Float (String)	Internal power electronics (IGBT / MOSFET) heatsink surface temperature.
`active_plimit`	`% or kW`	Float (String)	Active power curtailment or grid-compliance limit currently imposed.
`reactive_plimit`	`% or kVAR`	Float (String)	Reactive power compensation limit or Q-curve target currently commanded.
`WHDay / KWH_Day`	`Wh / kWh`	Float (String)	Cumulative energy generated by the inverter during the active day.
`WHTot`	`Wh / kWh`	Float (String)	Lifetime cumulative active generation energy recorded since commissioning.
`AC_CAP / DC_CAP`	`kW / kWp`	Float (String)	Nameplate AC nominal capacity and connected DC array capacity for this inverter unit.
`Day_PR`	`%`	Float (String)	Individual inverter unit daily performance ratio.
`RUN_HOURS`	`Hours`	Float (String)	Total lifetime operational grid-feeding operating service hours.

8. STRING MONITORING BOX (SMB) TELEMETRY (PATH: DATA → {TIMESTAMP} → SMB)

Granular DC combiner box telemetry providing individual string currents, array voltages, internal enclosure temperatures, and DC power aggregates for underperformance root-cause identification.

PARAMETER KEY	UNIT	DATA TYPE	ENGINEERING DESCRIPTION & CHANNEL PURPOSE
smb_type	String	String	String Monitoring Box hardware classification or internal telemetry mapping code.
suryalog_status	Status	Integer/String	RS485 Modbus communication health status of the SMB card (1 = Normal, 0 = Fault).
user_status	Status	Integer/String	Device operating mode and health alarm code flags.
smb_addon_data	Raw/JSON	String	Supplemental diagnostic information or vendor proprietary SMB register block.
smb_addon_active	Boolean	Integer/String	Status indicator for extended SMB diagnostic feature processing.
smb_data_active	Boolean	Integer/String	Indicates active real-time transmission of DC string current and voltage registers.
V	V	Float (String)	Instantaneous operating DC bus voltage at the combiner output busbar.
VMIN	V	Float (String)	Minimum DC string voltage recorded across all channels on this combiner.
VMAX	V	Float (String)	Maximum DC string voltage recorded across all channels on this combiner.
VAVG	V	Float (String)	Arithmetic mean average DC string voltage across all channels.
I1	A	Float (String)	Direct current (DC) measured on Channel 1 (String 1).
I2	A	Float (String)	Direct current (DC) measured on Channel 2 (String 2).
I3	A	Float (String)	Direct current (DC) measured on Channel 3 (String 3).
I4	A	Float (String)	Direct current (DC) measured on Channel 4 (String 4).
I5	A	Float (String)	Direct current (DC) measured on Channel 5 (String 5).
I6	A	Float (String)	Direct current (DC) measured on Channel 6 (String 6).
I7	A	Float (String)	Direct current (DC) measured on Channel 7 (String 7).
I8	A	Float (String)	Direct current (DC) measured on Channel 8 (String 8).
I9	A	Float (String)	Direct current (DC) measured on Channel 9 (String 9).
I10	A	Float (String)	Direct current (DC) measured on Channel 10 (String 10).
IMIN	A	Float (String)	Lowest individual DC string current recorded across all active channels.
IMAX	A	Float (String)	Peak individual DC string current recorded across all active channels.
IAVG	A	Float (String)	Mean average DC string current across all active conducting strings.
ITOT	A	Float (String)	Aggregated direct current (DC) summation across all strings in the combiner.
WTOT	W / kW	Float (String)	Total calculated direct current (DC) power generated by the combined strings.
TExt	°C	Float (String)	External ambient temperature probe at the SMB field enclosure mounting location.
TInt	°C	Float (String)	Internal enclosure electronics temperature (for thermal runaway monitoring).

9. STANDARD HTTP STATUS CODES & POSTMAN TROUBLESHOOTING

The API combines standard IETF HTTP response headers with Suryalog custom status descriptors (`result`, `cmsg`) within the JSON envelope.

HTTP CODE	STATUS MEANING	ROOT CAUSE / FAILURE CONDITION	CORRECTIVE ENGINEERING ACTION
`200 OK`	Request Successful	Server processed the request. Always verify payload `result` code.	Inspect the `result` property in the JSON body. If 0, ingest data.
`400 Bad Req`	Bad Request	Malformed JSON payload, invalid timestamp format, or missing required parameter.	Validate parameter names, ensure timestamps are 10 digits, and confirm valid JSON syntax.
`401 Unauth`	Unauthorized	Invalid, revoked, or missing cryptographic API key in `secret` parameter.	Verify the API secret provided in your tenant portal or request key renewal.
`403 Forbidden`	Forbidden Access	Secret key is authenticated, but does not have permission for the requested plant.	Contact your Suryalogix administrator to ensure the plant ID is bound to your key.
`404 Not Found`	Not Found	Endpoint URL misspelled, version path invalid, or plant ID non-existent.	Check dynamic `{version}` in URL and ensure target plant identifier exists.
`405 Method`	Method Not Allowed	Request dispatched using GET, PUT, or other disallowed HTTP verb.	Ensure client library explicitly issues an `HTTP POST` request.
`408 Timeout`	Request Timeout	Server timed out while aggregating requested range.	Narrow the query timestamp interval (`etime - stime`) to a shorter window.
`429 Rate Limit`	Too Many Requests	Rate limit quota exceeded by client application.	Introduce client rate limiting (max 1 req/sec recommended) and exponential backoff.
`500 Internal`	Internal Server Error	Unexpected exception during server-side database query execution.	Retry request using randomized exponential backoff; notify Suryalogix Support if persistent.
`502 Bad Gateway`	Bad Gateway	Upstream database or caching layer failed to respond to the API gateway.	Temporary cloud routing issue. Retry query in 30–60 seconds.
`503 Service Unavail`	Service Unavailable	Cloud platform undergoing scheduled maintenance or database scaling.	Pause polling operations and resume following the maintenance notification window.

TOP-LEVEL STATUS DIAGNOSTICS

Response Envelope Evaluation Rule: A status of `HTTP 200 OK` guarantees network connectivity, but does not guarantee data availability. Client ETL code must perform a two-step assertion:

```
if (http_status == 200 && response_json.result === 0) { /* Process response_json.data */ } else { log_error(response_json.cmsg); }
```

10. STANDARD OPERATING PROCEDURE (SOP) & IMPLEMENTATION BEST PRACTICES

- **Result Code Verification:** Always validate that `result === 0` before inspecting sub-nodes. If `result !== 0`, read `cmsg` to extract descriptive diagnostics (e.g., "No Data for Selected Time Window" or "Plant Offline").
- **Dynamic Endpoint Configuration:** Never hardcode API version numbers or domains directly inside business logic. Store base URLs and version tags (`{version}`) in external environment variables to facilitate non-disruptive platform upgrades.
- **Safe Numerical Type Conversion:** Because telemetry values (e.g., `W2_val`, `WT`, `VLL`) are transmitted as string representations in JSON to preserve decimal precision, client ETL engines must parse them into IEEE-754 double/float primitives prior to arithmetic processing.
- **Query Window Optimization:** Limit single data extraction queries (`etime - stime`) to a maximum window of 24 hours (86,400 seconds) when requesting high-frequency unaggregated logs. For historical backfilling, page requests sequentially in 12-hour chunks.
- **Synchronization with Logger Transmission Cadence:** Polling the API faster than the physical logger's transmission interval (typically 1, 5, or 15 minutes) creates duplicate data. Align client cron jobs with the plant-specific logging interval.
- **Zero-Padded String Null Checks:** When inverters or SMB channels are unpopulated or in nighttime sleep mode, registers may return `"0.000"`, `null`, or omit dormant channels. Integrate defensive null checks across all parsers.
- **Cryptographic Key Rotation:** Rotate the API `secret` key annually or whenever administrative personnel change. Key revocation and regeneration requests can be executed via the Suryalog administrative portal.

11. TECHNICAL SUPPORT & INCIDENT ESCALATION MATRIX

ESCALATION LEVEL	CONTACT & CHANNEL	SCOPE OF ASSISTANCE
Tier 1: API Support	`support@suryalogix.com`	Key generation, parameter clarification, endpoint availability inquiries.
Tier 2: Engineering & Analytics	`analytics@suryalogix.com`	Custom protocol mapping, high-frequency rate-limit adjustments, firmware issues.
Direct Portal Access	`https://cloud.suryalog.ae`	Plant configuration, live monitoring dashboards, user permission management.